

ZHARKENT, KazakhstanWMO: **368590**Lat: **44.1628N**Lon: **80.0458E**Elev: **2116**StdP: **13.61**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.4	1.5	-9.9	3.5	-2.5	-4.5	4.7	2.6	14.2	21.4	11.3	25.2	2.9	10	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	95.0	67.9	92.3	66.6	89.9	65.6	70.2	89.2	68.7	87.8	67.4	86.0	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	97.7	77.1	62.4	91.3	75.5	60.8	86.0	75.2	35.7	89.4	34.2	88.2	33.1	86.3	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	12.9	10.0	DB	-7.7	99.9	6.5	1.9	-12.4	101.2	-16.2	102.3	-19.9	103.4	-24.7	104.8	
(5)			WB	-8.2	74.2	6.5	2.1	-12.9	75.7	-16.6	76.9	-20.3	78.1	-24.9	79.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	20.7	27.0	43.0	56.7	65.9	73.4	76.3	74.1	65.7	53.1	38.8	25.4
	DBStd	20.86	9.28	9.27	9.50	8.30	6.76	5.27	4.78	5.41	6.79	8.00	8.78	8.42
	HDD50	3015	910	645	255	34	1	0	0	0	1	60	347	762
	HDD65	5931	1375	1065	683	263	71	6	1	5	74	376	785	1227
	CDD50	3673	0	0	37	237	494	702	816	748	472	155	12	0
	CDD65	1110	0	0	0	15	99	257	352	287	94	6	0	0
	CDH74	12588	0	0	9	286	1166	2748	3914	3274	1098	93	0	0
	CDH80	5385	0	0	0	68	377	1159	1864	1531	372	14	0	0
	WSAvg	4.1	3.0	3.5	4.7	5.6	5.5	4.8	4.2	4.0	3.9	3.7	3.4	3.0
	PrecAvg	7.1	0.4	0.4	0.5	0.8	0.7	1.1	0.9	0.4	0.3	0.6	0.7	0.6
(15) Precipitation	PrecMax	11.8	2.9	1.5	1.5	2.4	2.1	2.9	3.5	1.5	1.2	2.1	2.2	3.2
	PrecMin	3.4	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.3	0.4	0.3	0.4	0.6	0.5	0.8	0.8	0.4	0.3	0.6	0.5	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	53.2	74.5	86.7	91.7	95.1	99.4	97.6	92.1	82.0	67.3	48.7
		MCWB	37.0	42.1	52.8	59.4	62.7	66.5	69.7	69.8	65.7	58.6	49.8	39.3
	2%	DB	38.7	46.7	68.4	81.1	87.8	92.1	95.7	94.3	87.6	76.4	61.2	43.4
		MCWB	33.0	38.1	50.1	57.8	61.9	65.6	68.4	68.0	63.5	56.3	47.8	35.8
	5%	DB	35.5	42.5	63.6	76.8	84.3	89.7	92.7	91.5	84.6	71.7	55.5	39.4
		MCWB	31.6	35.9	48.4	56.1	60.6	65.0	67.4	66.9	62.3	54.4	45.3	33.8
	10%	DB	32.9	38.7	58.7	72.6	80.7	86.9	89.6	88.6	80.8	67.1	50.9	35.9
		MCWB	30.0	33.6	46.6	54.3	59.0	64.0	66.4	65.9	60.6	52.3	43.0	32.0
	0.4%	WB	37.6	42.5	54.8	62.0	66.6	70.1	73.4	72.4	68.2	60.7	52.9	40.8
		MCDB	42.6	50.6	69.9	78.9	85.9	88.2	92.2	92.4	87.1	77.7	62.5	47.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.3	39.1	51.8	59.4	63.7	68.1	70.8	69.8	65.1	57.7	48.9	36.9
		MCDB	37.3	45.6	65.2	77.3	82.3	86.5	88.9	90.4	84.5	73.2	58.6	41.9
	5%	WB	32.2	36.3	49.3	57.2	62.0	66.7	69.2	68.2	63.2	55.5	46.3	34.4
		MCDB	34.9	41.4	61.5	73.9	79.2	84.6	87.8	87.9	82.0	68.4	54.9	38.6
	10%	WB	30.0	34.1	47.1	55.4	60.5	65.4	67.7	66.7	61.3	53.5	43.6	32.1
		MCDB	33.0	38.2	57.6	69.7	76.8	82.5	86.4	85.6	78.6	64.9	50.2	35.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	16.9	19.6	22.9	23.3	22.7	23.4	25.2	25.6	22.8	17.8	16.3
		MCDBR	19.6	19.9	26.8	29.7	28.1	27.0	28.1	29.5	30.6	29.1	25.1	20.7
		MCWBR	14.2	12.4	12.8	13.6	11.0	9.9	10.9	13.1	14.6	13.9	13.8	14.2
		MCDBR	16.5	17.7	23.7	25.7	24.5	23.1	24.6	27.1	28.7	24.8	22.8	18.8
	5% WB	MCWBR	12.2	11.4	11.9	12.2	10.4	9.3	10.5	12.6	14.2	12.3	13.2	13.0
(40) Clear-Sky Solar Irradiance	taub		0.315	0.358	0.413	0.447	0.422	0.420	0.412	0.388	0.373	0.372	0.349	0.313
	taud		2.213	2.119	2.014	1.985	2.135	2.209	2.252	2.304	2.292	2.196	2.172	2.209
	Ebn at Noon		249	256	256	258	270	270	271	274	268	248	231	235
	Edh at Noon		29	38	48	54	48	44	42	39	36	35	30	26
(44) All-Sky Solar Radiation	RadAvg		649	955	1255	1621	2002	2071	2057	1915	1547	993	618	516
	RadStd		34	49	99	130	117	96	128	100	86	98	56	39

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days
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